Andi Zhang

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</> Summary

Assistant Professor at the University of Warwick, working on AI safety and the reliability of large language models. I develop probabilistic methods for adversarial robustness, out-of-distribution detection, and trustworthy generative models.

Work Experience

WMG, University of Warwick — Coventry, United Kingdom
Assistant Professor — Mar.2026 -

Department of Computer Science, University of Manchester — Manchester, United Kingdom
Postdoc Research Associate — Nov.2024 - Feb.2026

Education

Trinity College, University of Cambridge
PhD in Computer Science (Supervised by Dr. Damon Wischik) — Oct.2019 - Dec.2024
MASt in Mathematics (Part III of the Mathematical Tripos) — Oct.2018 - Jun.2019
MPhil in Advanced Computer Science — Oct.2017 - Jun.2018

School of Computer Science, University of Manchester
BSc (Hons) in Computer Science and Mathematics — Sep.2014 - Jun.2017

Internships

MediaTek Research — Cambridge, United Kingdom
Deep Learning Intern | Host: Dr. Alberto Bernacchia — Aug.2024 - Nov.2024
- Participated to a project in accelerating the sampling of diffusion models by proposing a new second-order estimation of the posterior, published in TMLR.

University College London — London, United Kingdom
Visitor | Hosts: Dr. David Barber and Dr. Mingtian Zhang — July.2023 - Oct.2023
- Worked on a new image generative model that based on Diffusion models.
- Worked on Amortized LoRA, a more principle and probabilistic way to generate LoRA parameters.

Kirin AI Group, Huawei — Cambridge, United Kingdom
Research Intern (Part-time) | Host: Mr. Arnau Raventos — Nov.2021 - Oct.2022
- Conducted a comprehensive evaluation of various traditional and AI-based frame interpolation methods, providing insights for the product department to leverage in their decision-making processes.
- Contributed to an AI global illumination project, successfully implemented various samplers within the Mitsuba 3 framework.

Noah's Ark Lab, Huawei — Shenzhen / Beijing, China
Research Intern | Hosts: Dr. Yitong Sun and Dr. Shifeng Zhang — Sep.2020 - Sep.2021
- Co-developed the local autoregressive model, which significantly advanced the state-of-the-art in Out-Of-Distribution (OOD) detection and lossless compression. The empirical evidence supporting this enhancement was recognized and published in NeurIPS 2021.
- Contributed to a computer vision project through researching deep learning uncertainty for active learning applications. The outcomes of this work were recognized and published in BMVC 2021.
- Contributed to an indoor positioning project leveraging WiFi signal strength. I specifically conducted research on Deep Gaussian Processes, leading our team to enhance indoor positioning accuracy. The successful results of this work were published in IPIN 2021.

University of Cambridge — Cambridge, United Kingdom
Course Supervisor (Teaching Assistant) — Oct.2019 - Dec.2024

- Supervised IA Algorithms (2020 Lent, 2023 Lent), IA Machine Learning in Real World (2020 Lent), IA Introduction to Probability (2020 Easter, 2021 Easter), IB Further Java (2020 Michaelmas), IB Data Science (2022 Michaelmas, 2023 Michaelmas), IB Further Graphics (2022 Michaelmas) and II Machine Learning and Bayesian Inference (2023 Lent)

MRC Biostatistics Unit, University of Cambridge — Cambridge, United Kingdom

Research Assistant | Hosts: Dr. Sofia Villar and Dr. David Robertson — Jul.2019 - Sep.2019

- Developed a simulation system to tackle the multi-armed bandit problem (MABP) in clinical trial settings and pioneered a unique test statistic for binary response MABP situations. Although the time constraints prevented this work from being formalized into a publication, it provided substantial practical and theoretical insight.

Tencent (WeChat AI) — Beijing, China

R & D Intern | Host: Dr. Jie Zhou — Jul.2018 - Sep.2018

- Engaged in comprehensive exploration and research in deep reinforcement learning and evolution strategies.

Amazon (Alexa) — Cambridge, United Kingdom

Applied Scientist Intern | Host: Dr. Emilio Monti — Jun.2017 - Sep.2017

- Employed the Seq2Seq model, the predecessor of the Transformer, with the Attention mechanism for semantic parsing tasks. Semantic parsing involves translating natural language into a structured internal language used for database or knowledge base querying.

Robotics Institute, Carnegie Mellon University — Pittsburgh, United States

Student Intern | Host: Dr. Stelian Coros — Jun.2016 - Sep.2016

- Contributed to the implementation of a general stance controller for 3D-printable robotic creatures. The controller's adaptability enables it to be used with any human-designed multi-foot robots, hence its designation as 'general'.

Publications (Last / Corresponding Author)

Let the Carrier Carry the Attack: Preserving the Subject in Adversarial Image Generation — Preprint
Linfeng Jiang, Steven McDonagh, Yuhang Chen, Xingyu Zhao, Siddartha Khastgir, **Andi Zhang**
https://arxiv.org/abs/2609.39723

From Expected Harmfulness to Likelihood: A Probabilistic Reformulation of Jailbreaking LLM Agents — Preprint
Juanyang Xu, Zheng Wang, Xingyu Zhao, Siddartha Khastgir, **Andi Zhang**
https://arxiv.org/abs/2610.09973

Hierarchical Orthogonal Residual Spread for Precise Massive Editing in LLMs — ICASSP 2026 Oral
Xiaojie Gu, Guangxu Chen, Yuheng Yang, Jingxin Han, **Andi Zhang**
In the 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)

FedMC: Federated Manifold Calibration — ICLR 2026
Yanbiao Ma, Wei Dai, Gaoyang Jiang, Wanyi Chen, Chenyue Zhou, Yiwei Zhang, Fei Luo, Junhao Wang, **Andi Zhang**
In the 14th International Conference on Learning Representations (ICLR)

Compositional Attribute Imbalance in Vision Datasets — AAAI 2026
Yanbiao Ma, Jiayi Chen, Wei Dai, Dong Zhao, Zeyu Zhang, Yuting Yang, Bowei Liu, Jiaxuan Zhao, **Andi Zhang**
In the 40th Annual AAAI Conference on Artificial Intelligence (AAAI)

Publications (First / Co-first Author)

Probabilistic Adversarial Training — Preprint
Andi Zhang, Xingyu Zhao, Siddartha Khastgir
https://arxiv.org/abs/2609.39798

Concept-based Adversarial Attack: a Probabilistic Perspective — ICLR 2026
Andi Zhang, Xuan Ding, Steven McDonagh, Samuel Kaski
In the 14th International Conference on Learning Representations (ICLR)

Your Finetuned Large Language Model is Already a Powerful OOD Detector — AISTATS 2025
Andi Zhang, Tim Z. Xiao, Weiyang Liu, Robert Bamler, Damon Wischik
In Proceedings of the 28th International Conference on Artificial Intelligence and Statistics

Constructing Semantics-Aware Adversarial Examples with a Probabilistic Perspective — NeurIPS 2024
Andi Zhang, Mingtian Zhang, Damon Wischik
In the 38th Annual Conference on Neural Information Processing Systems, 2024

SR-OOD: Out-of-Distribution Detection via Sample Repairing — Preprint
Rui Sun[*], **Andi Zhang**[*], Haiming Zhang[*], Yao Zhu, Ruimao Zhang, Zhen Li

[*]Equal contribution.

https://arxiv.org/abs/2305.18228

Falsehoods that ML researchers believe about OOD detection **NeurIPS 2022 Workshop**
Andi Zhang, Damon Wischik
In ML Safety Workshop, 36th Conference on Neural Information Processing Systems

Out-of-Distribution Detection with Class Ratio Estimation **NeurIPS 2022 Workshop**
Mingtian Zhang*, **Andi Zhang***, Tim Z. Xiao, Yitong Sun, Steven McDonagh
In ML Safety Workshop, 36th Conference on Neural Information Processing Systems

On the Out-of-distribution Generalization of Probabilistic Image Modelling **NeurIPS 2021**
Mingtian Zhang*, **Andi Zhang***, Steven McDonagh
In Neural Information Processing Systems annual meeting, 2021

Other Publications

Improved Denoising Diffusion Probabilistic Models with Efficient Non-Diagonal Covariance Modeling **TMLR**
Rui Xia*, Ayan Das*, Artem Artemev, **Andi Zhang**, Guillaume Hennequin, Alberto Bernacchia
In Transactions on Machine Learning Research (TMLR)

From Perception to Cognition: A Survey of Vision-Language Interactive Reasoning in Multimodal Large Language Models **AI Review**
Chenyue Zhou, Mingxuan Wang, ..., **Andi Zhang**, ... , Yike Guo
In Artificial Intelligence Review

Noise Diffusion for Enhancing Semantic Faithfulness in Text-to-Image Synthesis **CVPR 2025**
Boming Miao, Chunxiao Li, Xiaoxiao Wang, **Andi Zhang**, Rui Sun, ZiZhe Wang, Yao Zhu
In the IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)

Improving Probabilistic Diffusion Models With Optimal Diagonal Covariance Matching **ICLR 2025 Oral**
Zijing Ou*, Mingtian Zhang*, **Andi Zhang**, Tim Z. Xiao, Yingzhen Li, David Barber
In the 13th International Conference on Learning Representations (ICLR)

Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep **IPIN 2021**
Ran Guan, **Andi Zhang**, Mengchao Li, Yongliang Wang
In International Conference on Indoor Positioning and Indoor Navigation, 2021

Towards dynamic and scalable active learning with neural architecture adaption for object detection **BMVC 2021**
Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, **Andi Zhang**, Wei Zhang, Hongtao Lu, Chunjing Xu
In British Machine Vision Conference, 2021

Solipsistic Reinforcement Learning **ICLR 2021 Workshop**
Mingtian Zhang*, Peter Noel Hayes*, Tim Z. Xiao, **Andi Zhang**, David Barber
In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations, 2021

Academic Service

Area Chair:

- ICLR (2027)
- ICML (2027)

Reviewer:

- NeurIPS (2024, 2025, 2026)
- ICML (2025, 2026)
- ICLR (2025, 2026)
- AISTATS (2023, 2024, 2025, 2026, 2027)
- CVPR (2025, 2026)
- ICCV / ECCV (2025, 2026)
- ACM MM (2025)
- AAAI (2026, 2027)
- COLM (2026)